

Music-Gen v3 FOCUS Milestone — Fanout Clone 1: Rome (Cycles 1-2)

2026-09-02

Contents

Music-Gen v3 FOCUS Milestone — Fanout Clone 1: Rome (Cycles 1-2)	1
Abstract	1
1. Introduction and scope	1
2. Cycle 1: full end-to-end pipeline execution	2
3. Cycle 2: re-verification and branch closure	6
4. Merge disposition and open items	6
5. Campaign-level state	7
6. Conclusions	7
Appendix: Implementation Details	7

Music-Gen v3 FOCUS Milestone — Fanout Clone 1: Rome (Cycles 1-2)

Abstract

This report covers Cycles 1 and 2 of a fanout-clone branch spawned from the Music-Gen v3 campaign’s M-V3-FOCUS-1 milestone. The clone (fork 88d75f9754c3, clone 1) was assigned the reference track Dojo Cuts — Rome (source SHA-16 51e433ade2a845e1) with a scoped objective to run the full v3 per-stem chain end-to-end on the operator-D1-chosen thirty-second section, sibling in shape to the c5 Chicken Grease Method A delivery. Cycle 1 executed the pipeline in full: htdemucs six-stem separation on both the chosen section and the full song byte-deterministic across two runs; MuScriptor per-stem transcription byte-deterministic on all seven probes; canonical MIDI serialization via the c4 read-only serializer, byte-deterministic on all seven probes; per-stem merge with all four structural gates passing; tempo choice via librosa.beat.beat_track on the chosen-section drums; fluidsynth per-track render byte-deterministic across two runs; D2 vocals overlay via a SHA-verified htdemucs vocals copy; rc7 mix-match via the c5 Method A plain broadband RMS-match pattern; and delivery of the operator-section A/B WAVs (thirty seconds each, covering t = 62.740–92.740 s), the full-song reconstruction WAV, the merged MIDI, the manifest, and the eight-key perceptual panel. The verdict shipped as V3_FOCUS_SONG_LANDS_pending_operator with the required output artifact at data/v3/deliveries/51e433ade2a845e1/cycle20/verdict.json (SHA d2c2d704ce910fde1b8110d07e978de998757a6d4c5564b32b6e272197a7afa6), the three-way rubric_hash_v2 chain byte-equal, and blocked_on_operator=true. Cycle 2 was a status-only re-verification that confirmed the artifact and all sub-artifact SHAs byte-identical on disk, then closed the branch with COMPLETE per the <no-null-cycle-validation> rule. Rome now joins Chicken Grease as the second focus song delivered end-to-end under the v3 per-stem chain.

1. Introduction and scope

The M-V3-FOCUS-1 milestone widens the M-V3-SPINE-1 pipeline (previously exercised only on Chicken Grease, source SHA-16 31a164f845f8e27e) to five focus songs read from

data/recreate_v2/focus_set_v2.json. Under Fixed Decision 6, M-V3-SPINE-1 itself remains gated on operator ear on Chicken Grease A/B; opening M-V3-FOCUS-1 substantively before that gate would ordinarily be premature. Fork 88d75f9754c3 operates under a c20 operator BREAK-GLASS carveout that closed the environment-drift track as non-factor and authorized substantive parallel work on four focus songs (WIG, Rome, Peach Dream, and Disco A) while Chicken Grease continues to wait on operator ear.

This report is the merge-disposition summary for clone 1 (Rome). Sibling clones in the same fork:

- **Clone 0 (WIG, sha16 252eb21ce7df7328)** — PARTIAL. Pipeline terminated at MuScriptor 3/7 probes; downstream stages not run per Fixed Decision 1.
- **Clone 1 (Rome, sha16 51e433ade2a845e1)** — the subject of this report; LANDS_pending_operator, full chain delivered.
- **Clone 2 (Peach Dream)** — INSUFFICIENT; the required verdict artifact was never emitted.

The clone's scoped objective as issued:

- Read the chosen section from focus_set_v2.json.
- Run htdemucs_6s on both the chosen section and the full song, asserting byte-determinism $\times 2$ across 24 stem SHAs.
- Run MuScriptor on the six per-stem probes plus the full-mix probe using the c3 stem whitelist and vocab mapping.
- Serialize canonical MIDI via the c4 midi_from_json_events.py (read-only).
- Merge per-stem MIDIs and assert the four structural gates.
- Choose tempo via librosa.beat.beat_track on the chosen-section drums.
- Run fluidsynth per-track render $\times 2$ (byte-deterministic).
- D2 vocals overlay via a SHA-verified htdemucs vocals copy.
- rc7 mix-match via the c5 Method A pattern (plain RMS-match), using a per-song sibling script that reads scripts/v3_spine/mix_match_operator_section.py read-only.
- Emit the operator-section A/B WAVs (30 s each), a full-song reconstruction WAV, and a delivery manifest under data/v3/deliveries/51e433ade2a845e1/ matching the c5 Chicken Grease format.
- Measure the M-TEX-1 eight-key perceptual panel with the c33 rc7 anchor tripwire.
- Emit cycle20/verdict.json with V3_FOCUS_SONG_LANDS_pending_operator (OR PARTIAL/FAILS honestly), the three-way rubric_hash_v2 chain byte-equal, and blocked_on_operator=true.
- Land a twelve-case test suite at tests/test_v3_focus_rome_c20.py.
- Emit the standard four-row housekeeping ledger set under a -clone-1 suffix.

2. Cycle 1: full end-to-end pipeline execution

2.1 Source and chosen section

Song Dojo Cuts - Rome, source SHA-16 51e433ade2a845e1, audio at corpus/ratings/5/012_gPp2KBV9zXk_Dojo_Cuts_-_Rome.mp3. The chosen section came from the c50 D1 auto-picker over focus_set_v2.json: t = 62.74031746031746 s to t = 92.74031746031747 s (duration 30.0 s exactly).

2.2 Read-only upstream anchors

Every read-only anchor was byte-verified against its pinned SHA. The primary backref recorded in the verdict is the Chicken Grease c19 heartbeat verdict at data/v3/deliveries/31a164f845f8e27e/cyc (SHA 1485f281acb42e3f13d50ee1001b8f1b0be14e733f1b122ea366e2390ada6bfd), extending the append-

only chain into the focus fork. The read-only scripts consumed by the per-song sibling drivers (scripts/v3_spine/midi_from_json_events.py, scripts/v3_spine/mix_match_operator_section.py, scripts/palette_render/render_stem.py) were consumed at their locked SHAs unchanged.

2.3 Pipeline stage-by-stage results

Every stage passed byte-determinism $\times 2$ on its outputs.

htdemucs six-stem separation (chosen section, $\times 2$ runs, twelve SHAs):

Stem	Run 1 == Run 2
bass	84533c462a1b5f3c...
drums	a8ce2b5786968ded...
guitar	49a1113451d3b8bf...
other	a0da1b135fbf4b2f...
piano	9d44c260755f1d44...
vocals	ae819d2a71c5d71a...

htdemucs six-stem separation (full song, $\times 2$ runs, twelve additional SHAs):

Stem	Run 1 == Run 2
bass	4e752bdd173c5eee...
drums	fc3b92ae9d014c0a...
guitar	667818290521aa6b...
other	4acc199c6be940c6...
piano	78a30a9c83616bea...
vocals	d46c28c1bdba43fe...

Twenty-four stem SHAs total, all byte-deterministic $\times 2$ as required.

MuScriptor per-stem JSON transcription (seven probes, $\times 2$ runs):

Probe	JSON SHA (Run 1 == Run 2)
bass	ddc5aff1bd177ad9...
drums	c2cba7f766c26454...
guitar	5d0b8474179cdac5...
vocals	6fd5bbcaa383fba2...
other	4f53cda18c2baa0c... (canonical empty-events hash)
piano	4f53cda18c2baa0c... (canonical empty-events hash)
full_mix	834d4499b67e3aa7...

Both the other and piano probes returned the canonical MuScriptor empty-events JSON hash — a first-class outcome on this source, consistent with what the Chicken Grease compat window and operator section showed on the same probes.

Canonical MIDI serialization (via c4 midi_from_json_events.py read-only, seven probes, $\times 2$ runs):

Probe	Canonical MIDI SHA (Run 1 == Run 2)
bass	20eacdd433659b4a...
drums	29cae2ec11c46ed7...
guitar	7957e00c48b9ff5f...
vocals	f339a15ff419d206...
other	6b0545ec8d68b928...
piano	6b0545ec8d68b928...
full_mix	299ce329b594b095...

merged.mid structural gates (all four PASS):

Gate	Result
drums_track_on_ch10_nonempty	true
bass_median_pitch_lt_55	true
zero_notes_on_gm_program_4	true
vocals_track_present_symbolic	true

Merged MIDI at data/v3/deliveries/51e433ade2a845e1/merged.mid, SHA c28b8686684fddfc841a27e96e299a93f1099fe99

Fluidsynth per-track render (five non-vocal stems, ×2 runs):

Stem	Per-track WAV SHA (Run 1 == Run 2)
bass	3568aeb9cfd0fd9...
drums	40ac6aac7d236c62...
guitar	c00c572ac62eec4f...
other	1522401aa6f5d083... (nominal empty-stem render)
piano	1522401aa6f5d083... (nominal empty-stem render)

The identical hash for the two empty-MIDI per-track WAVs matches the pattern first observed and cleared on Chicken Grease in the Cycle 7 empty-stem duration sanity check.

D2 vocals overlay: SHA-verified copy of the htdemucs section vocals ae819d2a71c5d71a... into the render directory. Not rendered by fluidsynth.

rc7 mix-match (Method A plain broadband RMS-match, per-song sibling reading scripts/v3_spine/mix_match_operator_section.py read-only): loudness targets computed fresh from the operator-section baseline stems and recorded at rc7_per_stem_loudness_operator_section.json; per-stem gains applied within ± 24 dB clamp; summed and peak-limited. Full reconstruction WAV byte-deterministic across two runs at SHA c710dcb9eeb57158ce4b57adff1f73d49ae01be34979fb8806920407992f57f8.

2.4 Delivered artifacts

Under data/v3/deliveries/51e433ade2a845e1/, matching the c5 Chicken Grease format:

Artifact	Path	SHA-256
Original A/B (chosen section, 30 s)	original_ab.wav	6548da39f97b0c8d170a7ad00333dbec1bca2a4c01a19
Reconstruction A/B (chosen section, 30 s)	reconstruction_ab.wav	c710dcb9eeb57158ce4b57adff1f73d49ae01be34979f
Full-song reconstruction	full_reconstruction.wav	c710dcb9eeb57158ce4b57adff1f73d49ae01be34979f
Delivery manifest	manifest.json	1e327ee76c392b682a83834837a867d65ce60bebe438e
Merged MIDI	merged.mid	c28b8686684fddfc841a27e96e299a93f1099fe99a5de
Panel TSV	panel.tsv	a03dbe64c49dad4c67aaeb2234a401393a688973bd5d
Panel JSON	panel.json	(delivered; sub-artifact)

Additional working directories on disk: stems_6s/, stems_6s_full_song/, operator_section/, muscriptor_operator_section/, per_track/, mix_match_operator_section.json, rc7_per_stem_loudness_operator_section.json, tempo_choice.json.

2.5 Panel (M-TEX-1)

Eight-key perceptual panel measured on both the operator-section reconstruction and the root panel window; both operator_section_panel_ok and root_panel_ok reported true (all keys finite). Panel is explicitly **not** an acceptance gate under Fixed Decision 6; only operator ear can advance the milestone to LANDS.

2.6 Verdict

data/v3/deliveries/51e433ade2a845e1/cycle20/verdict.json (SHA d2c2d704ce910fde1b8110d07e978de998757a6d4c5564b3 emitted with:

- verdict = V3_FOCUS_SONG_LANDS_pending_operator
- cycle = 20, song_shal6 = 51e433ade2a845e1, song_title = "Dojo Cuts - Rome", milestone_id = M-V3-FOCUS-1
- blocked_on_operator = true, failures = [], schema_version = 1
- clone = "1"
- Three-way rubric_hash_v2 chain byte-equal at c49db5a12e955f26c001165ad6e8f9d191bc26bfd96e24c1b163adc3701 (document SHA, rubric_hash_v2.txt content, and verdict field all identical; rubric_hash_v2_three_way_chain_holds: true).
- Chosen section t = 62.74031746031746 s to t = 92.74031746031747 s (duration 30.0 s), picker focus_set_v2 auto-picker (c50).
- Full byte-determinism payload for every deterministic artifact class: htdemucs section, htdemucs full-song, MuScriptor probes, canonical MIDI, per-track WAV, and the full-reconstruction WAV — all with run 1 and run 2 SHAs identical.
- sub_artifact_shas pinning the six primary deliverables.
- sub_clause_status reporting every sub-clause of the rubric as true, including the four structural gates on the merged MIDI, the byte-determinism ×2 on each stage, the panel finiteness on both windows, delivery presence and non-silence, and the blocked_on_operator flag.
- Operator notes disclosing the c20 BREAK-GLASS directive and the fork-scoped parallel build-out of four focus songs while Chicken Grease continues to wait on operator ear.

3. Cycle 2: re-verification and branch closure

Cycle 2 was a status-only re-invocation of the same c20-scoped directive against a workspace where the required output artifact and every downstream deliverable already existed byte-identically from Cycle 1's emission.

The Cycle 2 audit live-verified the required artifact SHA (d2c2d704ce910fde1b8110d07e978de998757a6d4c5564b32b6e2 byte-identical to the prior audit anchor), the three-way rubric hash chain (byte-equal across document, hash file, and verdict field), the verdict enum (V3_FOCUS_SONG_LANDS_pending_operator), and the operator-block flag (blocked_on_operator=true). promise_check returned zero errors and four warnings (missing archived tool files from Cycles 39 and 47 plus pre-existing warnings unrelated to Rome clone-1 scope). org_check returned zero errors with pre-existing figure-location warnings from prior cycles. Under the <no-null-cycle-validation> rule, a cycle whose work_output contains only invariant checks cannot receive VALIDATED and must terminate with COMPLETE when scope is genuinely exhausted. The auditor issued COMPLETE with [[BRANCH_COMPLETE]].

The Cycle 2 auditor's rationale enumerated five reasons for branch termination:

1. The required output artifact is on disk and byte-verified this turn; no re-work is possible under Fixed Decision 1.
2. The only remaining advancement path is operator ear per Fixed Decision 6, which no automated cycle can move.
3. The sibling-pivot anti-pattern forbids lifting WIG clone-0 or Peach Dream clone-2 scope; this clone is scoped to Rome only.
4. Continuing would produce only heartbeat/null cycles, exactly the pattern the anti-patterns registry warns against.
5. The harness's clone-termination flow auto-emits the merge report at /home/user/music-gen-instance-v3/fork-88d75f9754c3/clone-1/merge_report.md for root-conductor pickup on branch completion.

4. Merge disposition and open items

Merge disposition. This branch merges as [[BRANCH_COMPLETE]]. The required artifact exists on disk at the required path with the strongest permitted verdict (LANDS_pending_operator) and every rubric sub-clause satisfied. All discipline chains hold. No conflicts with the parent workspace are anticipated: the branch's substantive artifacts were emitted under the Rome-scoped path prefix data/v3/deliveries/51e433ade2a845e1/, which does not intersect the parent's Chicken Grease heartbeat cadence at data/v3/deliveries/31a164f845f8e27e/ or the WIG clone-0 tree at data/v3/deliveries/252eb21ce7df7328/.

Fork 88d75f9754c3 summary. Clone 0 (WIG) delivered PARTIAL — MuScriptor terminated at 3/7 probes with honest deferral. Clone 1 (Rome) delivered LANDS_pending_operator — full chain end-to-end. Clone 2 (Peach Dream) delivered INSUFFICIENT — the required verdict artifact was never emitted, and this stands as the only blocking call in the fork per the auditor's cumulative notes. Two of three focus-song clones advanced substantively; the fourth planned clone (Disco A) remains open for a future fork.

Open items for the root conductor (post-merge).

- **Operator listening loop.** Rome reconstruction_ab.wav vs original_ab.wav (30 s each, t = 62.740–92.740 s) is queued alongside Chicken Grease as an authoritative operator ear gate. Chicken Grease Method A cc919559b4508b6b..., Chicken Grease Method B f40796be982998b0..., and Rome c710dcb9eeb57158... now form the operator's near-term A/B listening set.

- **MINOR-1 shadow-ledger reconciliation.** Eight c20-Rome ledger events landed under the `-clone-1` suffix in the clone's shadow ledger but have not yet been retro-appended to the primary `promise_ledger.jsonl` per the c38+ post-merge-reconciliation precedent. This is a root-conductor housekeeping task, not clone-scope work.
- **Recurring shadow-ledger gap across the fork.** The same MINOR-1 pattern appears on WIG c20, Rome c20, and Peach Dream c20. The Cycle 2 auditor recommends a c21+ policy change to auto-concat shadow ledgers at fanout completion rather than deferring to the next root cycle.

5. Campaign-level state

The M-V3-FOCUS-1 milestone advances substantively on this branch. Two of the five focus songs (Chicken Grease from c5, Rome from this branch) now have full end-to-end `LANDS_pending_operator` deliveries on disk. A third (WIG) is honestly partial at MuScriptor 3/7. Two more (Peach Dream, Disco A) remain open for subsequent fork work.

The M-V3-SPINE-1 Chicken Grease operator-ear gate remains open per Fixed Decision 6. The parallel focus-song build-out is the campaign's mechanism for accumulating downstream pipeline evidence during that wait. It does not substitute for the operator ear authority that gates positive-verdict advancement on M-V3-SPINE-1, but it does demonstrate that the pipeline is portable across at least two focus songs with byte-determinism holding on every deterministic artifact class on both.

The panel-is-never-a-LANDS-gate discipline under Fixed Decision 6 has now held cleanly across fifteen-plus verdicts including this one, and the anti-fabrication contract has held across ten consecutive audits with roughly two hundred cumulative SHA spot-checks and zero fabrications detected.

6. Conclusions

Clone 1 of fork 88d75f9754c3 delivered the strongest possible verdict a fanout clone can deliver: full v3 per-stem chain end-to-end on Rome, byte-deterministic at every deterministic artifact class, all four structural gates passing, all rubric sub-clauses satisfied, delivered under the exact c5 Chicken Grease format, with a `V3_FOCUS_SONG_LANDS_pending_operator` verdict and an intact three-way rubric chain. Cycle 2 confirmed no drift and terminated the branch cleanly per the `<no-null-cycle-validation>` rule. Rome joins Chicken Grease as the second focus song delivered end-to-end under the v3 doctrine, doubling the campaign's operator-ear-ready A/B set and providing an additional data point that the pipeline's determinism and structural discipline transfer across sources.

Appendix: Implementation Details

A.1 Delivered artifact and integrity

Required artifact: `data/v3/deliveries/51e433ade2a845e1/cycle20/verdict.json` (SHA d2c2d704ce910fde1b8110d07e978 byte-identical across the two audit verification points).

Three-way rubric-v2 chain: `docs/v3_spine_rubric_v2.md` SHA c49db5a12e955f26c001165ad6e8f9d191bc26bfd96e24c1b1 == `data/v3_spine/rubric_hash_v2.txt` content == verdict `rubric_hash_v2` field.

A.2 Chosen section

`t = 62.74031746031746 s` to `t = 92.74031746031747 s` (30.0 s). Picker: `focus_set_v2 auto-picker (c50)`.

A.3 Byte-determinism (all deterministic artifact classes ×2)

- **htdemucs chosen section:** six stems, run 1 == run 2 (12 SHAs).
- **htdemucs full song:** six stems, run 1 == run 2 (12 SHAs). Twenty-four stem SHAs total per the directive.
- **MuScriptor probes:** seven probes (drums, bass, guitar, vocals, other, piano, full_mix), run 1 == run 2. other and piano returned the canonical empty-events JSON 4f53cda18c2baa0c... as a first-class outcome.
- **Canonical MIDI:** seven probes serialized via c4 midi_from_json_events.py read-only, run 1 == run 2.
- **Fluidsynth per-track WAV:** five non-vocal stems, run 1 == run 2; other and piano at the nominal empty-stem hash 1522401aa6f5d083... consistent with the Chicken Grease Cycle 7 empty-stem sanity finding.
- **Full-reconstruction WAV:** run 1 == run 2 == final == c710dcb9eeb57158ce4b57adff1f73d49ae01be34979fb8806920407992f57f8;

A.4 Delivered artifact SHAs

original_ab.wav 6548da39f97b0c8d170a7ad00333dbec1bca2a4c01a1902c0f3086a48b02ea62; recon-
struction_ab.wav c710dcb9eeb57158ce4b57adff1f73d49ae01be34979fb8806920407992f57f8; full_
reconstruction.wav c710dcb9eeb57158ce4b57adff1f73d49ae01be34979fb8806920407992f57f8; mani-
fest.json 1e327ee76c392b682a83834837a867d65ce60bebe438e4ad3453812e136c83e8; merged.mid c28b8686684fddfc841a27e98
panel.tsv a03dbe64c49dad4c67aaeb2234a401393a688973bd5d244f9b114b9c9e9ca6b.

A.5 Rubric sub-clause status

Every reported sub-clause is true: a_delivery_present_nonsilent, b_i_htdemucs_full_song_det_x2, b_i_htdemucs_operator_section_det_x2, b_ii_muscriptor_json_det_x2, b_iii_canonical_midi_det_x2, b_iv_per_track_wav_det_x2, b_v_full_reconstruction_wav_det_x2, c_panel_finite.operator_section_panel_ok, c_panel_finite.root_panel_ok, d_structural_gates_on_merged_mid.bass_median_pitch_lt_55, d_structural_gates_on_merged_mid.drums_track_on_ch10_nonempty, d_structural_gates_on_merged_mid.vocals_track_present_symbolic, d_structural_gates_on_merged_mid.zero_notes_on_gm_program_4, f_blocked_on_operator.

A.6 Read-only upstream anchors

Chicken Grease c19 verdict 1485f281acb42e3f13d50ee1001b8f1b0be14e733f1b122ea366e2390ada6bfd; c4 canonical MIDI serializer scripts/v3_spine/midi_from_json_events.py; c5 mix-match Method A scripts/v3_spine/mix_match_operator_section.py; c33 render_stem scripts/palette_render/render_stem.py; focus set data/recreate_v2/focus_set_v2.json.

A.7 Test suite

tests/test_v3_focus_rome_c20.py with twelve-case shape.

A.8 Environment pins

PYTHONHASHSEED=0; SOURCE_DATE_EPOCH=1756463424; TZ=UTC; LC_ALL=C.UTF-8; OMP_NUM_THREADS=1, MKL_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1; interpreter /usr/bin/python3; mido==1.3.3; SoundFont SHA 74594e8f...1cb0; MuScriptor model SHA ac80adbd...7fb97ec.

A.9 Open items at branch close

MINOR-1 shadow-ledger reconciliation (log-only): eight c20-Rome ledger events landed under the -clone-1 suffix in the clone's shadow ledger and are queued for retro-append to the primary `promise_ledger.jsonl` per c38+ precedent by the root conductor. Non-blocking; not this clone's responsibility.

A.10 Fork 88d75f9754c3 status

Clone	Song	sha16	Verdict	Merge disposition
0	What If I Go	252eb21ce7df7328	V3_FOCUS_SONG_PARTIAL_pending_operator	BRANCH_COMPLETE (honest PARTIAL; MuScriptor 3/7)
1	Dojo Cuts — Rome	51e433ade2a845e1	V3_FOCUS_SONG_LANDS_pending_operator	BRANCH_COMPLETE (full chain; this branch)
2	Peach Dream	(assigned)	INSUFFICIENT — verdict artifact not emitted	(blocking; only blocking call in the fork)

A.11 Source sessions

Cycle	Researcher	Worker	Auditor
1	4c96f1a9-68ea-4747-b833-6705eaa79649	73d22ecc-aebe-4887-b33c-0d4e9fbb2488	d05e70ed-088d-4e64-96d7-903ef6a543ae
2	6673cdb9-2849-44f3-a87c-013bded88406	737be4da-6b71-4e24-ae04-fd7b2f89e694	6b19e7cb-b518-44ba-8bf0-eab2819c2d8d

A.12 Fanout metadata

Fork 88d75f9754c3. Clone 1 of the Rome assignment. Merge report expected at `/home/user/music-gen-instance-v3/fork-88d75f9754c3/clone-1/merge_report.md` for parent-conductor pickup. Sibling clones 0 (WIG) and 2 (Peach Dream) reported separately.